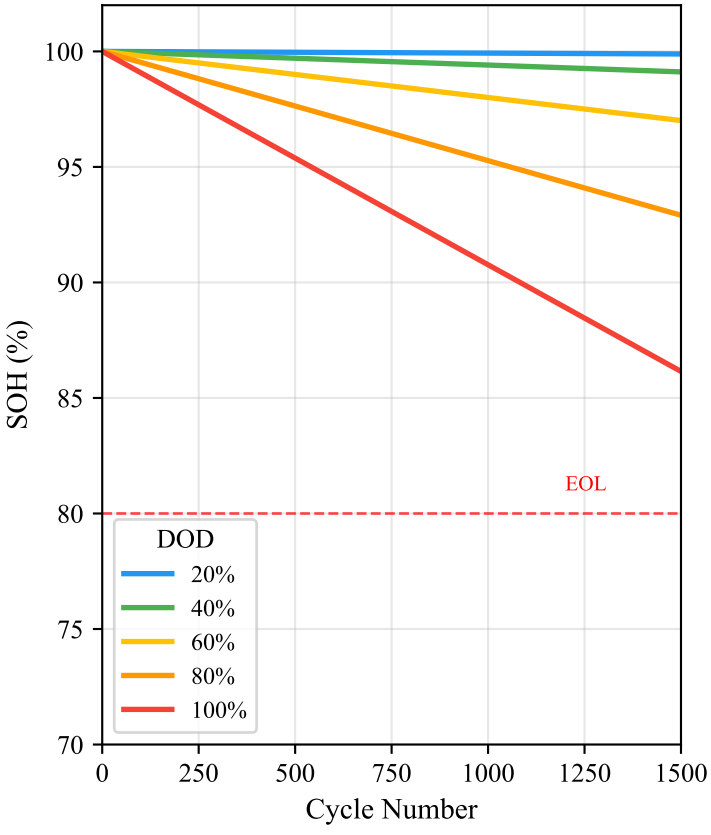

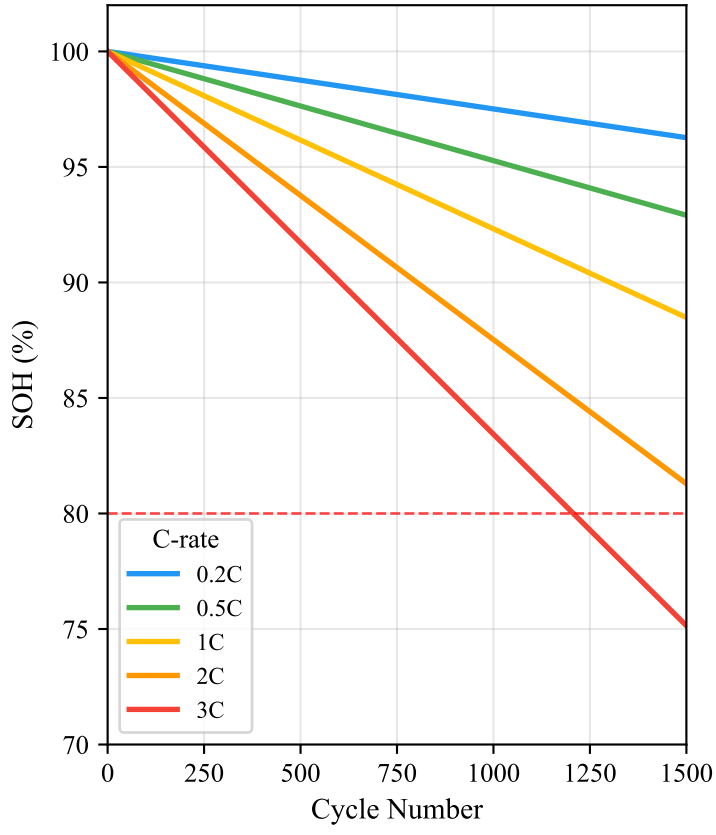

Cycle Aging Model

$$\frac{dSOH_{cyc}}{dt} = -k_{cyc} \cdot DOD^{\gamma} \cdot \left(\frac{I}{I_{nom}}\right)^{\alpha} \cdot |I|$$

$$k_{cyc}(T) = A_{cyc} \cdot \exp\left(-\frac{E_{a,cyc}}{R \cdot T}\right)$$

Parameters :

$k_{cyc} = 5 \times 10^{-5}$ Ah^{-1}

$\gamma = 2.0$ (DOD exponent)

$\alpha = 0.7$ (C-rate exponent)

$I_{nom} = 3.0$ A (1C rate)

Key Insights :

- Deep discharge accelerates wear
- High C-rate causes Li plating
- Mechanical stress from cycling